COMPETITIVE PROGRAMMING ROADMAP

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Zero to ICPC Regional Level Complete Concept Syllabus

Level 1: Basics & Fundamentals

Basic I/O & Operators

- Data Types, Fast I/O Tricks
- Conditional Statements (`if-else`, `switch`)
- Loops (`for`, `while`, `do-while`)

Arrays & Strings (Basic)

- 1D/2D Arrays, Multidimensional Storage
- String Operations, `char` array vs `std::string`

Functions & Pointers

- Pass by Value / Pass by Reference
- Memory Addresses & Pointers Basics
- Recursion Fundamentals

Basic Mathematics

- Modulo Arithmetic Basics
- Parity Checks, Divisibility Rules
- Prime Identification (Naive $O(\sqrt{N})$)
- GCD & LCM (Euclidean Algorithm)

Time & Space Complexity

- Big-O, Big-Ω, Big-Θ Notations
- Operations per Second Estimation (10^8 ops $\approx$ 1 sec)

Standard Template Library (STL)

- Sequence: `vector`, `pair`, `deque`, `list`
- Adapters: `stack`, `queue`, `priority_queue`
- Associative: `set`, `multiset`, `map`, `multimap`
- Unordered: `unordered_set`, `unordered_map`
- Essential Algorithms: `sort`, `reverse`, `min_element`, `max_element`, `lower_bound`, `upper_bound`, `next_permutation`, `__builtin_popcount`

Level 2: Elementary Techniques

Prefix Sum & Difference Array

- 1D Prefix Sum & Range Queries
- 2D Prefix Sum (Grid Submatrix Sum)
- String / Frequency Prefix Sum
- Difference Array (Range Updates in $O(1)$)

Two Pointers Technique

- Opposite Direction Pointers (e.g., 2-SUM)
- Same Direction Pointers (e.g., Subarray Sum)
- Fast and Slow Pointers (Cycle Detection)

Sliding Window Technique

- Fixed-size Sliding Window
- Variable-size / Dynamic Sliding Window

Binary Search & Ternary Search

- Binary Search on Sorted Arrays
- Binary Search on Answer Space (Monotonic Predicates)
- Fractional Binary Search (Floating point precision)
- Ternary Search (Unimodal Functions / Optimization)

Sorting Techniques & Applications

- Custom Comparators & Struct Sorting
- Coordinate Compression
- Merge Sort & Inversion Count

Greedy Algorithms

- Activity Selection & Interval Scheduling
- Fractional Knapsack

Level 3: Number Theory & Mathematics

Number Theory

- Sieve of Eratosthenes (10^7 limit)
- Segmented Sieve
- Prime Factorization ($O(\sqrt{N})$ & Sieve-based $O(\log N)$)
- Euler's Totient Function $\phi(N)$
- Modular Arithmetic (Addition, Subtraction, Multiplication)
- Modular Exponentiation (Binary Exponentiation)
- Modular Inverse (Fermat's Little Theorem, Ext. GCD)
- Chinese Remainder Theorem (CRT)

Combinatorics

- Permutations & Combinations ($nCr \pmod M$)
- Pascal's Triangle
- Pigeonhole Principle
- Inclusion-Exclusion Principle (In-Ex)
- Catalan Numbers
- Probability & Expected Value Basics

Matrix Exponentiation

- Fast Fibonacci & Linear Recurrences in $O(k^3 \log N)$

Level 4: Graph Theory

Graph Representation

- Adjacency Matrix, Adjacency List, Edge List

Graph Traversals

- Breadth-First Search (BFS)
- Depth-First Search (DFS)
- Connected Components & Bipartite Checking

Trees

- Tree Traversals, Tree Diameter & Center
- Lowest Common Ancestor (LCA) via Binary Lifting
- Euler Tour Technique (Flattening Trees)

Shortest Path Algorithms

- Dijkstra's Algorithm ($O((V+E)\log V)$)
- Bellman-Ford Algorithm (Negative Cycles)
- Floyd-Warshall Algorithm ($O(V^3)$ All-Pairs)
- 0-1 BFS

Disjoint Set Union (DSU)

- Path Compression & Union by Rank/Size
- Small to Large Merging (DSU on Tree)

Minimum Spanning Tree (MST)

- Kruskal's Algorithm
- Prim's Algorithm

Topological Sort

- Kahn's Algorithm (BFS-based)
- DFS-based Topological Sort
- Cycle Detection in Directed Graphs

Level 5: Dynamic Programming (DP)

Basic DP Concepts

- Memoization (Top-down) vs Tabulation (Bottom-up)
- State Representation & Transitions

Standard Classical DP

- 0/1 Knapsack & Unbounded Knapsack
- Longest Common Subsequence (LCS)
- Longest Increasing Subsequence (LIS) in $O(N \log N)$
- Coin Change & Edit Distance
- Matrix Chain Multiplication (MCM)

Advanced DP Types

- Grid DP
- Bitmask DP
- Digit DP
- DP on Trees
- Interval / Range DP
- Convex Hull Trick / Li Chao Tree Optimization
- Divide & Conquer Optimization
- Sum over Subsets (SOS) DP

Level 6: Advanced Data Structures

Segment Tree

- Point Update, Range Query
- Lazy Propagation (Range Updates)
- Merge Sort Tree / Dynamic Segment Tree
- Persistent Segment Tree

Fenwick Tree (BIT)

- Point Update, Range Query
- Range Update, Point Query
- 2D Fenwick Tree

Trie (Prefix Tree)

- String Insertion & Search
- Maximum XOR Pair / Bitwise Trie

Square Root Decomposition

- Block Decomposition
- Mo's Algorithm (Offline Range Queries)
- Mo's Algorithm on Trees

Sparse Table

- Range Minimum/Maximum Query (RMQ) in $O(1)$

Others

- Policy-Based Data Structures (C++ PBDS `policy_tree`)
- Treap / Implicit Treap

Level 7: String Algorithms & Advanced

Advanced String Algorithms

- String Hashing (Single & Double Hashing)
- KMP Algorithm (Prefix Function)
- Z-Algorithm
- Manacher's Algorithm (Palindromic Substrings)
- Aho-Corasick Automaton
- Suffix Array & LCP Array
- Suffix Automaton

Advanced Graph Theory

- Bridges & Articulation Points (Tarjan's)
- Strongly Connected Components (SCC)
- Euler Path & Euler Circuit
- 2-SAT Problem
- Max Flow (Dinic's Algorithm)
- Min Cut & Bipartite Matching (Hopcroft-Karp)
- Heavy-Light Decomposition (HLD)
- Centroid Decomposition

Computational Geometry

- Vector Operations (Cross & Dot Products)
- Convex Hull (Monotone Chain)
- Shoelace Formula (Polygon Area)
- Line Segment Intersection

Game Theory

- Nim Game, Sprague-Grundy Theorem, MEX

Level 8: Expert / ICPC Regional

- Fast Fourier Transform (FFT) / NTT
- Segment Tree Beats
- Wavelet Tree
- Parallel Binary Search
- Meet in the Middle
- Offline Queries with DSU Rollback
- Pollard's Rho Algorithm